

Liam Arzola

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EDUCATION

Ph.D. Computer Science

09/2024 – ongoing

University of California, San Diego - advised by Prof. Alex Snoeren and Prof. Amy Ousterhout

GPA: 3.95. Researching thread runtimes for CPU on-chip accelerators (i.e. Intel's DSA, IAA, and QAT).

B.A. Computer Science

08/2023

Cornell University

EXPERIENCE

Research Intern

06/26 – 09/26

Microsoft Research / Azure Systems Research Group

Researching hierarchical KV caching efficiency for LLM agent workflows.

Software Engineer

01/2024 – 08/2024

Microsoft

Worked in Azure's Core OS group. Contributed to VMBus and OpenVMM/OpenHCL

(<https://github.com/microsoft/openvmm> ), a modular, cross platform virtual machine monitor. (C, Rust)

Research Intern

06/2023 – 12/2023

The Max Planck Institute for Software Systems, Operating Systems Group, Prof. Antoine Kaufmann

Worked with Prof. Antoine Kaufmann on host network stack research.

- Worked on extending TAS, an optimized userspace TCP stack, by adding hardware offload and scatter/gather support.
- Worked on Virtuoso, a novel host networking stack for virtual machines and containers for low overhead performance isolation. (<https://arxiv.org/abs/2309.14016>).

Undergraduate Researcher

10/2022 – 08/2023

Cornell University, Theory Meets Practice Lab, Prof. Lorenzo Alvisi

Worked on Pesto, a byzantine fault tolerant relational database. I designed our relational frontend engine, implemented popular online transaction processing benchmarks (TPCC/Auctionmark), and integrated CockroachDB as a baseline system. (C++)

Software Engineer Intern

05/2022 – 08/2022

Microsoft

Worked on Azure Storage's Capacity Management team on designing and implementing a leader based consensus library (Raft) for fault tolerant load balancing. (C#, .NET).

Software Engineer Intern

06/2021 – 08/2021

Dropbox

Designed and implemented sync progress aggregation for backups. (Rust)

PUBLICATIONS

Pesto: Cooking up high performance BFT Queries

Accepted to The 31st Symposium on Operating Systems Principles (SOSP 2025)

Florian Suri-Payer (Cornell University), Neil Giridharan (UC Berkeley), **Liam Arzola** (UC San Diego), Shir Cohen Gahtan (Cornell University), Lorenzo Alvisi (Cornell University), Natacha Crooks (UC Berkeley).

Tail Contagion: Sub-microsecond Time Protection in Shared Software Network Datapaths

Under Submission

Matheus Stolet (MPI-SWS), **Liam Arzola** (UC San Diego), Simon Peter (University of Washington), Antoine Kaufmann (MPI-SWS).

HACKATHON AWARDS

1st Place Startup Prize at Miami Hack Week

2021

HfO Backend Capital

2nd Place at the Miami Bitcoin Hackathon

2018

1st Place at the Matrix & Visa IoT and AI Hackathon

2017

SKILLS

- Rust, C, C++, Terraform, SQL, Python, DPDK, gdb, perf, Wireshark, TCP